I chose Task A and fine-tuned *whisper-tiny* and *whisper-small* on German Common Voice (CV). I picked Whisper over a wav2vec2 checkpoint mainly for contamination reasons: most German wav2vec2 models on HF (Hugging Face) were already fine-tuned on CV, so "held-out" test data could sit inside the baseline. Whisper was not trained on CV specifically, and it's unlikely (though not provable) that it was part of Whisper's undisclosed training web-scrape. Running two sizes of the same family let me test a hypothesis: "Gains from fine-tuning shrink as the base model gets stronger". **With fine-tuning, *tiny* went from 46.9% to 33.7% WER, *small* from 15.3% to 13.1% WER**, both statistically real (paired bootstrap, neither CI (confidence interval) includes zero), so the hypothesis held. Full table, CIs and normalization rules are in the README.

My key data decision was to use the official CV *train/dev/test* splits unmodified because they guarantee no speaker and no sentence appears in more than one split, which I verified programmatically rather than trusting it directly. I then created fixed subsets with committed audio-clip IDs so that every run uses identical data. The test set was touched exactly twice per model: once before fine-tuning, once after, with identical decoding and normalization. Moreover, CV left HF in late 2025, and instead of using one of the unofficial mirrors, I downloaded the official release from Mozilla because I treated this assignment like a real company one where "where did this data come from and what are we allowed to do with it" matters a lot. So I also read the terms, and saw re-sharing the data is explicitly forbidden. That's why my repository ships no audio, just raw model outputs and reference texts, which reproduce all my results on a CPU.

The thing I changed after seeing it fail was my training schedule: planned 1500 steps for both models. *Tiny's* WER on dev stopped improving after step 1000, so checkpoint selection kept the step-1000 model. *Small*'s first attempt crashed near the end, but it already showed dev WER flat from step 500 to 1000 with validation loss rising, so I did the rerun with 500 steps (to not waste GPU). Something that did not help was number normalization (spelling out digits because num2words reads the German years wrong). It shifts WER only 0.3 points, because apparently only 43 of 1000 test references contain digits, and the fine-tuned models learned to spell numbers out anyway.

The number I do not fully trust is *tiny*'s 13.2-point improvement. Its bootstrap CI came out suspiciously wide, so I inspected per-clip errors: on two clips the base model got stuck in repetition loops, producing 440 and 177 errors against references of 10 and 13 words. With loop clips excluded, the improvement is 8.3 points with a tight CI. Fine-tuning removed both loops but introduced one new, milder one, so it reduced this failure mode without eliminating it. In a clinical setting this is the failure I would worry about most: the model invents fluent text instead of failing visibly, and average error rates do not surface it unless you inspect more closely. The weakest part overall is domain: I proved gains on the CV dataset, which is Wikipedia-style sentences read out loud; not spontaneous, conversational, clinical speech. So I would not assume that the improvement transfers to the clinical domain without measuring it.